1. $15 = r + a + b$
   $12 = \frac{1}{2}r + 2b + a$
   $r = 2a + 2b$
   $15 = 2(a + b) + a + b$
   $15 = 3a + 3b$
   $12 = 0.5(2a + 2b) + 2b + a$
   $12 = a + b + 2b + a$
   $12 = 2a + 3b$

$15 = 3a + 3b$
$12 = 2a + 3b$
$3 = 1a$
$3 = a$

$12 = 6 + 3b$
$6 = 3b$
$b = 2$

$r = 6 + 4$
$r = 10$

2. $16 = x + y + z$
   $x = y + z$
   $3z = x + 1$
   $16 = 3z - 1 + y + z$
   $16 = 4z - 1 + y$
   $17 = 4z + y$

$16 = 2z + 2y$
$8 = z + y$

$17 = 4z + y$
$8 = z + y$
$9 = 3z$
$3 = z$

$17 = 12 + y$
$5 = y$

$16 = x + 5 + 3$
$16 = x + 8$
$8 = x$

3. x

$25 = n + d + q$
$490 = 5n + 10d + 25q$
$d = 2n - 1$
$25 = n + (2n - 1) + q$
$25 = 3n - 1 + q$
$26 = 3n + q$

$490 = 5n + 10(2n - 1) + 25q$
$490 = 5n + 20n - 1 + 25q$
$500 = 25n + 25q$
$20 = n + q$
$26 = 3n + q$
$-6 = -2n$
$3 = n$

$20 = 3 + q$
$17 = q$
$d = 6 - 1$
$d = 5$

4. x

   1. $t = b * (2)^d$

      $b = start\ value$

      $d = days$

      $t = total\ infected\ people$

   2. $\sim f(y) = 20 \cdot (2)^{5.645}$

5. $(8 + 2x)(4 + 2x) = 165$

$32 + 16x + 8x + 4x^2 = 165$

$32 + 24x + 4x^2 = 165$

$24x + 4x^2 - 133 = 0$

$x = \frac{-24 \pm \sqrt{24^2 - 16(-133)}}{8}$

$x = \frac{-24 \pm \sqrt{576 + 2128}}{8}$

$x = \frac{-24 \pm 52}{8}$

no neg distance so:

$x = \frac{28}{8}$

$x = 3.5$